CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (ME) Examination

December 2013

ME 713.01 Advanced Materials

Date: 30.12.2013, Monday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 State different definitions of nanotechnology. Classify nanostructured materials with examples and figures. [07]

Q - 2 (a) Explain opacity and translucency for insulators. [07]

(b) Explain evaporation condensation growth mechanism. [07]

OR

Q - 2 (a) Explain refraction phenomenon in detail. [07]

(b) Write short note on Chemical Vapor Deposition (CVD) method. [07]

Q - 3 Attempt **any two** [14]

(a) Explain in detail various factors affecting contact angle measurements.

(b) Explain in detail the effect of surface roughness on contact angle.

(c) Explain in detail the contact angle hysteresis and their causes.

SECTION – II

Q - 4 Explain thermal stress in homogeneous and bilayer structures. [06]

Q - 5 (a) Derive the Stoney formula. [08]

(b) How would you analyze the Load vs. Displacement curves obtained during nanoindentation testing? [06]

OR

Q – 5 (a) Define the types of stress in thin film. Explain thermal stress in three-layer structures. [08]

(b) What is Nanohardness? Explicate the method to determine nanohardness. [06]

Q - 6 Write short note on **any three** [15]

(a) Cantilever beams method.

(b) Circular plate method.

(c) Applications of advanced nanostructured materials such as (i) friction and wear nanolaminates and superlattices (ii) cutting tools (iii) biomedical.

(d) X-Ray Diffraction stress measurement technique.

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